

EXAM 7 – SPRING 2014

7. (2 points)

Identify four common data issues associated with applying an over-dispersed Poisson bootstrap model and provide one adjustment for each.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 7	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: A8: Identify data issues and related model adjustments for reserving models.
SAMPLE ANSWERS	
<i>Sample Answer 1</i> ○ Negative incremental value: limit incremental losses to zero ○ Missing values/incomplete data: estimate missing values using surrounding values ○ Heteroscedasticity: stratified sampling is accomplished by organizing the development periods by group with homogeneous variance within each group and then sampling with replacement only from the residuals in each group ○ Exposures that have changed dramatically over the years: modify data to get pure premiums and multiply the residuals by the exposures by year after the process variance step <i>Sample Answer 2</i> ○ Non-zero sum of residuals: add a single constant to all residuals so that the sum of the adjusted residuals is zero ○ Outliers/extreme values: exclude outliers from the average age-to-age factors and residual calculations, but re-sample the corresponding incremental when simulating triangles ○ Heteroecthesious data/misshapen data/partial year/interim evaluation dates: project future incremental values before applying model; need to annualize then de-annualize results ○ Lack of residuals to sample from/lack of extreme residuals: need to parameterize a distribution from which to sample from <i>The following data issues/adjustments were also accepted:</i> ○ Negative incremental value: remove the row from the triangle if it is causing extreme results and doesn't improve the parameterization of the model; use $-\ln(\text{abs}(q(w,d)))$ for $q(w,d) < 0$, when total of all incremental values in a development column is positive; when column(s) sum to a negative value, add the absolute value of the largest negative (either among the sums of development columns or incremental values in the triangle) to every incremental value in the triangle, solve the GLM, and reduce each fitted incremental value by the largest negative; when simulating from a negative incremental value $m(w,d)$ using the gamma distribution:	

SAMPLE ANSWERS AND EXAMINER'S REPORT

use $\text{Gamma}(\text{abs}(m(w,d)), \phi \times \text{abs}(m(w,d))) + 2 \times m(w,d)$, since this will have mean $m(w,d)$ while remaining skewed to the right; discussion in 4.1.1 contains other alternatives

- Missing values/incomplete data: modify loss development factors to exclude the missing value (in which case there will not be a corresponding residual for this missing value)
- Outliers/extreme values: remove them and deal with them like missing values; exclude outliers from the average age-to-age factors and residual calculations, but re-sample the corresponding incremental when simulating triangles
- Heteroscedasticity: sort development periods into groups with homogeneous variances, multiply each residual in each group by a hetero-adjustment factor ($\text{max}(\text{st.dev.}(r_i))/\text{st.dev.}(r_i)$), sample with replacement among all residuals, and divide each residual by the hetero-adjustment factor when the residuals are resampled

EXAMINER'S REPORT

Candidates were expected to provide four data issues and briefly identify a solution for each. While the solution was required to be complete and correct, the candidate was not required to recall specific formulas or equations. As an example, if the candidate's adjustment for negative incremental values required subtracting the largest negative, the candidate was also required to mention that this adjustment needs to be removed after solving the GLM. Credit would not be given if the candidate's adjustment required adding the largest negative with no mention of using the absolute value.

Most candidates were successful in identifying at least 3 data issues with appropriate adjustments (whether it was directly from the list or a description of one of the items). For example, several candidates did not use the term "heteroecthesious data", but they were able to give an accurate description of the term. Per CAS exam rules, only the first 4 identified responses were graded in instances where candidates identified more than 4 items.

Common errors:

- Candidates were not given credit for providing multiple aspects for what essentially is the same data issue. For instance, if candidates listed multiple data issues associated with Negative Incremental Values only the first of these responses was accepted.
- Some candidates mismatched appropriate adjustments between two different data issues. For instance, the paper mentions that it is common for outliers to form the first row of the data triangle and that removing the first row is an appropriate solution in this instance. However, this was sometimes listed as an adjustment for missing values, which isn't mentioned in the paper (the paper lists an example in which the oldest diagonals are missing). Credit for identifying the data issue was still given in these cases.
- Incomplete responses were sometimes given. Often an adjustment required applying a constant or factor, but the solution was missing the removal of this constant or factor after fitting or sampling. Credit for identifying the data issue was still given in these cases.